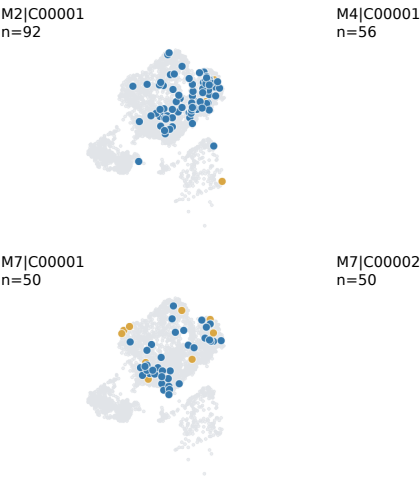


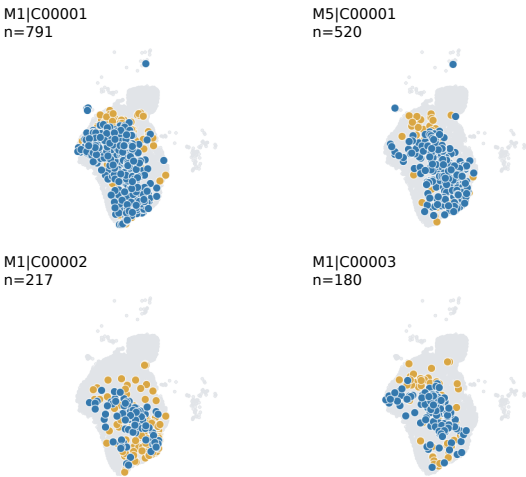
Supplementary Figure 2 | Model-antigen clone relationships and orthogonal GC labels

**A** NP-OVA clones spanning division gates



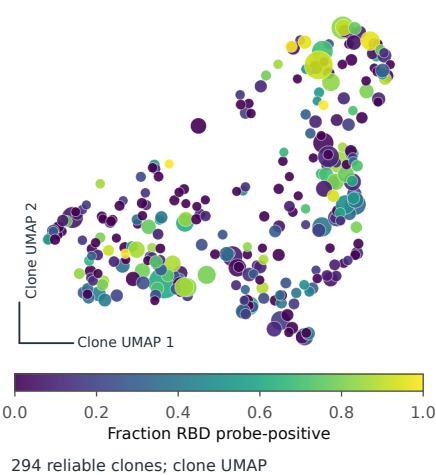
Each inset shows one BCR-defined clone on the same cell UMAP.

**B** RBD clones spanning division gates

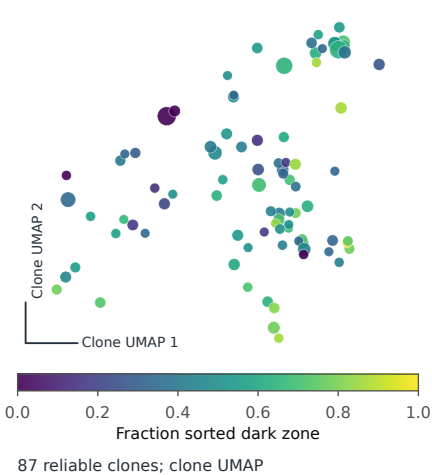


Each inset shows one BCR-defined clone on the same cell UMAP.

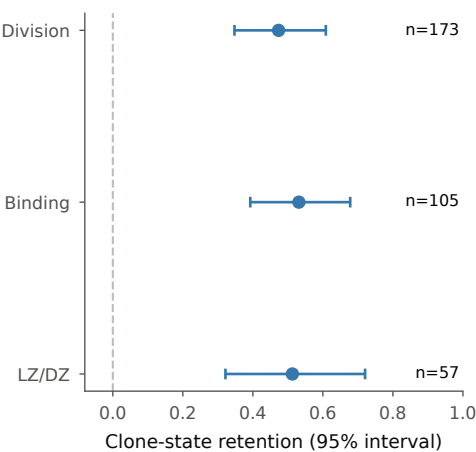
**C** RBD protein: antigen binding



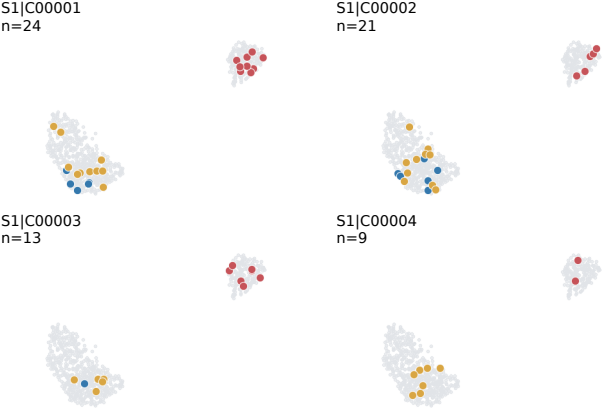
**D** RBD mRNA: DZ occupancy



**E** State across GC gates



**F** Directly measured NP-OVA output compartments



Each inset shows one BCR-defined clone on the same cell UMAP.

**G** Library-stratified reporter associations

